EXPERIMENT 12: KNOWLEDGE GRAPH TRAVERSAL USING NETWORKX

Aim:

To find a path between two entities in a knowledge graph using shortest-path search.

Software Required: Python 3.x and NetworkX.

Install NetworkX using:

pip install networkx

Procedure / Algorithm:

1. Create a graph of related concepts.
2. Specify the starting and target entities.
3. Use shortest_path() to find a connection.
4. Display the path or a message if no path exists.

Program:

```
graph = {
   "Arun": ["Python", "College"],
   "Python": ["Arun", "Programming"],
   "Programming": ["Python", "Computer Science"],
   "Computer Science": ["Programming"],
   "College": ["Arun"]
}


start = input("Enter starting entity: ")
target = input("Enter target entity: ")


queue = [[start]]
visited = []


found = False
```

```
while queue:
    path = queue.pop(0)
    node = path[-1]

    if node == target:
        print("Connection Path:", " -> ".join(path))
        found = True
        break

    if node not in visited:
        visited.append(node)

        for neighbour in graph.get(node, []):
            if neighbour not in visited:
                queue.append(path + [neighbour])

if not found:
    print("No path found between the given entities.")
```

Sample Input:

Enter starting entity: Arun

Enter target entity: Computer Science

Expected Output:

```
=== RESTART: C:/Users/saimr/AppData/Local/Programs/Python/Python312/KE-12.py ===
Enter starting entity: MRUDULA
Enter target entity: CSE
No path found between the given entities.
```

Result:

Thus, the program found a path connecting two entities in the knowledge graph.